

Bigger PINNs Are Needed for Noisy PDE Training

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This presentation is based on a recent paper we are waiting to get published and was co-authored by

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A fundamental puzzle in ML is to understand the role of model size in determining trainability for a specific task. We make quantitative progress towards understanding this for neural nets solving non-linear PDEs - a core aspect of AI for science.

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- Physics-Informed Neural Networks (PINNs) provide an alternative numerical method that provably does not have curse of dimensionality for Kolmogorov PDEs (Theorem 3.3 of Mishra 2021, Lagaris, Likas, and Fotiadis 1998; De Ryck and Mishra 2024),
- This is done by introducing the PDE as a loss constraint into the training process of a neural nets.

Given a spatial domain $[-B, B]^n \times [0, T]$, a function $u(\mathbf{x}, t) : \mathbb{R}^n \times \mathbb{R}^+ \rightarrow \mathbb{R}$ is said to satisfy the Hamilton-Jacobi-Bellman (HJB) PDE if for a given function $g(\mathbf{x})$:

$$\begin{aligned}\frac{\partial u}{\partial t} - \nabla_{\mathbf{x}}^2 u + \|\nabla_{\mathbf{x}} u\|_2^2 &= 0 \\ u(\mathbf{x}, 0) &= g(\mathbf{x})\end{aligned}$$

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HJB has become a typical benchmark for PINNs (Nakamura-Zimmerer et al. (2020), Borovykh et al. (2022)) and has been used for control problems ranging over robotics, aeronautics, finance, etc (Bando, & Yamakawa, 2010, Prakash et al., 2022, Witte, & Reisinger, 2011).

Empirical Loss function of PINNs for HJB

Definition

The empirical PINN loss function for HJB, with regularization parameters $\lambda_0, \lambda_s > 0$, (for enforcing the initial condition and supervision data condition respectively), on a model h_w is:

$$\hat{R}(h_w) := \underbrace{\frac{1}{N_r} \sum_{i=1}^{N_r} \left(\partial_t h_w(\mathbf{x}_{ri}, t_{ti}) - \nabla^2 h_w(\mathbf{x}_{ri}, t_{ri}) + \|\nabla h_w(\mathbf{x}_{ri}, t_{ti})\|_2^2 \right)^2}_{\text{PDE Residual Loss}} + \underbrace{\frac{\lambda_0}{N_0} \sum_{j=1}^{N_0} (h_w(\mathbf{x}_{0j}, 0) - g(\mathbf{x}_{0j}))^2}_{\text{Initial Condition Loss}} + \underbrace{\frac{\lambda_s}{N_s} \sum_{k=1}^{N_s} (h_w(\mathbf{x}_{sk}, t_{sk}) - y_{sk})^2}_{\text{Supervised Loss With Noisy Solution Samples}}$$

Where N_r, N_0, N_s represents the number of points being tested for the residual, initial condition, and noisy supervision loss respectively

Numerical experiments for HJB and Taylor-Green

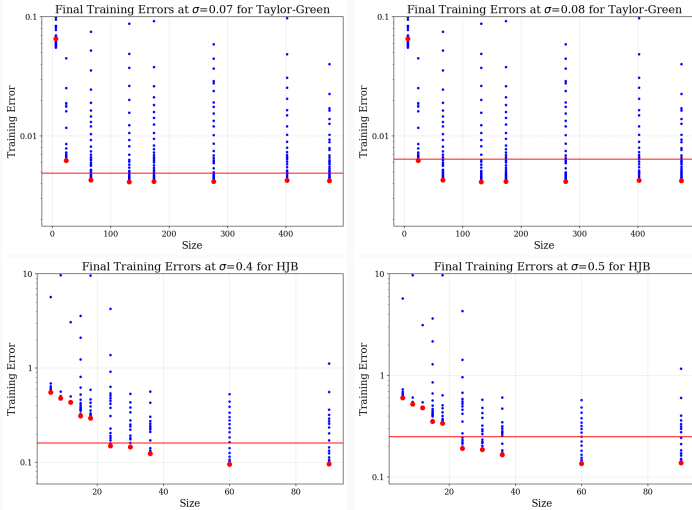
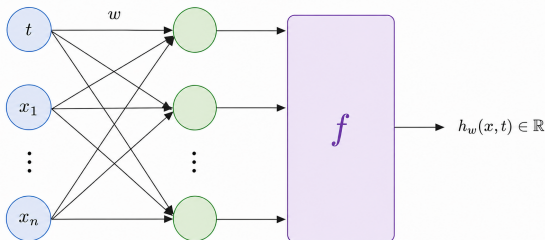


Figure 1: Training errors vs size for both the HJB and the Taylor-Green Solutions of the Navier-Stokes PDE, both showing similar behavior

Assumptions



For our model h_w , there is a fixed non-linear function $f : \mathbb{R}^k \rightarrow \mathbb{R}$ such that

$$\mathbb{R}^n \times \mathbb{R} \ni (x, t) \mapsto h_w(x, t) = f(W_1 x + w_2 t) \in \mathbb{R},$$

$$\text{where } w = (W_1, w_2) \text{ with } W \in \mathbb{R}^{k \times n}, w_2 \in \mathbb{R}^k.$$

W_1 and w_2 are the trainable parameters. This could be a two-layer MLP with the second layer frozen during training.

Theorem (Informal Theorem)

If a predictor can achieve an empirical risk below the variance of the label noise σ^2 , then necessarily there is a lower threshold on the number of trainable parameters of the PINN, d_N , required to achieve that empirical risk. Specifically, this lower bound is

$$d_N \log d_N \gtrsim N_s$$

where N_s is the number of samples.

Proof Structure

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Imposing a high probability for the existence of a good network results in the desired constraint.

Step 1

Lemma (Low Noise Correlation Implies Low Error)

Define the random variables $z_i = y_{si} - \mathbb{E}[y_{si} | (x_{si}, t_{si})]$. Let $\mathcal{P}$ be any class of parameterized functions $p_{\mathbf{w}} : \mathbb{R}^{n+1} \rightarrow \mathbb{R}$ for which $\hat{R}$ can be defined. Then for all $\eta, \mu > 0$,

$$\mathbb{P}\left(\exists p_{\mathbf{w}} \in \mathcal{P} \mid \hat{R}(p_{\mathbf{w}}) \leq \sigma^2 - \mu\eta\right) \leq 2 \exp\left(-\frac{N_s \eta^2}{288 M^4}\right) \\ + \mathbb{P}\left(\exists p_{\mathbf{w}} \in \mathcal{P} \mid \frac{1}{N_s} \sum_{i=1}^{N_s} p_{\mathbf{w}}(\mathbf{x}_{si}, t_{si}) z_i \geq S\right),$$

where $S = \frac{1}{2\lambda_s} (\mu\eta - \sigma^2) + \left(\frac{1}{2}\sigma^2 - \frac{1}{4}\eta\right)$.

Lemma (Correlation Bound)

The following probabilistic bound holds:

$$\begin{aligned} & \mathbb{P} \left(\exists h_{\mathbf{w}_{\frac{\eta}{2}}} \in \mathcal{H}_{\eta} \left| \frac{1}{N_s} \sum_{i=1}^{N_s} h_{\mathbf{w}_{\frac{\eta}{2}}}(\mathbf{x}_{si}, t_{si}) z_i \geq S \right. \right) \\ & \leq 2 \left(\frac{2W \sqrt{d_N}}{\eta} \right)^{d_N} \exp \left\{ \frac{-N_s S^2}{2 \cdot (8MC_1)^2} \right\} + 2 \exp \left\{ \frac{-N_s S^2}{32M^2 C_1^2} \right\} \end{aligned}$$

where $S = \frac{1}{2\lambda_s} (\mu\eta - \sigma^2) + (\frac{1}{2}\sigma^2 - \frac{1}{4}\eta)$ and $\mathcal{H}_{\eta}$ represents the finite cover of η separation in the weight space of the neural net function space.

Step 3

Lemma (Perturbation bound for PINN empirical risk for HJB)

Remembering the definition of the PINN loss $\hat{R}$, we have

$$\hat{R}(h_{\mathbf{w}_{\eta/2}}) \leq \hat{R}(h_{\mathbf{w}}) + a_1\eta + a_2\eta^2 + a_3\eta^3 + a_4\eta^4 \quad \forall \eta > 0, \quad (1)$$

This Lemma results from a thorough calculation specific to the HJB PINN loss, and so generalization to other PDEs would require a new version of this lemma.

The nonlinear terms of η result from only the nonlinear term in the HJB equation.

Perturbation bound constants

$$a_1 = \lambda_s C_2 V(n) (C_1 + M) + \lambda_0 C_2 \sqrt{n} B (C_1 + G) + b_1$$

$$a_2 = \frac{1}{4} (C_3 n k + (W C_3 V(n) + C_2)^2) (W C_2 (1 + W C_2) + C_3 n k W^2) \\ + \left((W C_3 V(n) + C_2) \left(\frac{1}{2} + W C_2 \right) + C_3 n k W \right) \\ \times \left(C_2 \left(\frac{1}{2} + W (C_3 V(n) + C_2^2 + 1) \right) + W \left(\frac{1}{2} C_3 V(n) + C_3 n k \right) \right)$$

$$a_3 = \frac{1}{4} (C_3 n k + W C_3 V(n) + C_2) \left((W C_3 V(n) + C_2) \left(\frac{1}{2} + W C_2 \right) + C_3 n k W \right) \\ + \frac{1}{4} (C_3 n k + (W C_3 V(n) + C_2)^2) \left(C_2 \left(\frac{1}{2} + W (C_3 V(n) + C_2^2 + 1) \right) \right) \\ + W \left(\frac{1}{2} C_3 V(n) + C_3 n k \right)$$

$$a_4 = \frac{1}{16} (C_3 n k + W C_3 V(n) + C_2) (C_3 n k + (W C_3 V(n) + C_2)^2).$$

Theorem

There exists a constant $\alpha > 0$ such that for all sufficiently small $\eta > 0$, and for all $\delta \in (0, 1)$, the following holds: If with probability at least $1 - \delta$ (with respect to the sampling of the supervision data), there exists $h_w \in \mathcal{H}$ such that

$$\hat{R}(h_w) \leq \sigma^2 - s\eta, \quad (2)$$

then the size of the model (d_N) is lower bounded by the number of supervision samples (N_s):

$$d_N \cdot \left(\ln(d_N) + \ln\left(\frac{4W^2}{\eta^2}\right) \right) \geq \frac{N_s \eta^2}{144M^4} - 2 \ln\left(\frac{4}{1-\delta}\right), \quad (3)$$

Combining the Low Noise Correlation Implies Low Error Lemma and the Correlation Bound Lemma results in:

$$\begin{aligned} & \mathbb{P}\left(\exists h_{\mathbf{w}_{\frac{\eta}{2}}} \in \mathcal{H}_{\eta} | \hat{R}(h_{\mathbf{w}_{\frac{\eta}{2}}}) \leq \sigma^2 - s\eta\right) \\ & \leq 2 \exp\left(-\frac{N_s \eta^2}{288 M^4}\right) + 2 \left(\frac{2W\sqrt{d_N}}{\eta}\right)^{d_N} \exp\left\{\frac{-N_s S^2}{2 \cdot (8MC_1)^2}\right\} \\ & + 2 \exp\left\{\frac{-N_s S^2}{32 M^2 C_1^2}\right\}, \end{aligned}$$

Using the Perturbation Lemma to convert the previous bound from the finite cover space to the full space to gives us:

$$\begin{aligned} \mathbb{P} \left(\exists h_{\mathbf{w}} \in \mathcal{H} | \hat{R}(h_{\mathbf{w}}) \leq \sigma^2 - s\eta - (a_1\eta + a_2\eta^2 + a_3\eta^3 + a_4\eta^4) \right) \\ \leq 2 \exp \left(-\frac{N_s \eta^2}{288M^4} \right) + 2 \left(\frac{2W\sqrt{d_N}}{\eta} \right)^{d_N} \exp \left\{ \frac{-N_s S^2}{2(8MC_1)^2} \right\} \end{aligned} \quad (4)$$

$$+ 2 \exp \left\{ \frac{-N_s S^2}{32(MC_1)^2} \right\}. \quad (5)$$

We lower-bound the probability by $1 - \delta$ and do some algebraic manipulation to derive our main theorem

- Is this relationship between N_s and d_N is also sufficient to guarantee low PINN risk?
- Can this be extended to systems with vector-valued solutions (e.g., the Navier–Stokes equations)? **This is essentially down to finding a perturbation lemma for Navier-Stokes.**
- Can this be extended to deeper and/or unbounded networks?